

System assessment Report

i.MX8M Plus Platform

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Chapter 1

Introduction

This document provides a baseline system assessment of the i.MX8M Plus application processor platform. Its purpose is to document the current hardware accessibility state as observed through runtime probing.

The following chapters follow the hardware hierarchy: compute and AI cores, memory and security, inter-core communication, multimedia and display, audio, networking and TSN, low-speed peripherals, power and clocks, and debug/trace.

Within each chapter, findings are detailed based on shell commands used to probe the system. to assess the current configuration, and assess the capabilities that this platform can perform.

Chapter 2

Compute & AI Cores

This chapter presents the actual system state of the i.MX8M Plus platform based on runtime probing. It covers the Cortex-A53 quad-core cluster, NEON SIMD capabilities, GPU, NPU, VPU, PMU counters, and a readiness assessment of the system to be used in the context of the project.

2.1 Application CPU : Cortex-A53 MPCore

The running system reports a homogeneous quad-core ARM Cortex-A53 cluster.

Property	Observed value
Architecture	aarch64 (32/64-bit capable)
Vendor / Model	ARM Cortex-A53 r0p4
CPU cores	4 (online mask 0-3)
Max frequency	1800 MHz (nominal)
Current frequency	1600 MHz (all cores)
Min frequency	1200 MHz (all cores)
Scaling governor	ondemand
L1 cache	128 KiB instruction + 128 KiB data (4 instances)
L2 cache	512 KiB (shared)
Flags	fp asimd evtstrm aes pmull sha1 sha2 crc32 cpuid

Note: No SVE (Scalable Vector Extensions) is present, which is expected for Cortex-A53. The `asimd` flag confirms NEON support. WFE and SGX are not exposed in `/proc/cpuinfo` flags; WFE/SEV are ARMv8 architectural instructions present by definition and not enumerated as `cpuinfo` features.

2.2 Memory Overview

System memory as reported by `/proc/meminfo` and zone information. Values are point-in-time readings and will vary between runs.

Parameter	Value
MemTotal	1.6 GB (1 627 172 kB)

MemFree	1.2 GB (1 248 464 kB)
MemAvailable	1.4 GB (1 426 548 kB)
CMA total	448 MB (458 752 kB)
CMA free	447 MB (457 888 kB)
Hugepage size	2 MB (Hugepages allocated: 0)
NUMA nodes	1 (node0, CPUs 0-3)

The large CMA pool (448 MB, DMA zone) indicates readiness for multimedia zero-copy buffers (ISP, VPU).

2.3 NEON / SIMD Capabilities

NEON is confirmed available in hardware. SVE is not supported on this architecture.

- **Flags in /proc/cpuinfo:** `fp asimd evtstrm aes pmull sha1 sha2 crc32 cpuid`
: confirmed on all four cores.
- **SVE:** not present. `/proc/sys/abi/sve_default_vector_length` absent. Expected behaviour : Cortex-A53 implements ARMv8.0; SVE requires ARMv8.2+.
- **Compiler NEON test:** gcc not installed; compile-time verification could not be performed. NEON availability is confirmed at the hardware level via CPU flags only.

2.4 Graphics Processing Unit : GC7000UL

The Vivante GC7000UL GPU DRM nodes are present but the kernel driver is not loaded and the userspace stack is absent.

Component	Status
DRM nodes	<code>/dev/dri/card0</code> , <code>/dev/dri/renderD128</code> present
GPU sysfs	<code>/sys/class/drm/</code> enumerated (symbolic link to <code>40000000.mix_gpu_ml</code>); <code>/sys/bus/platform/drivers/vivante/</code> absent
GPU kernel module	No module loaded (<code>galcore</code> , <code>etnaviv</code> , <code>vivante</code> absent from <code>lsmod</code>)
GPU device tree	<code>mix_gpu_ml@40000000</code> node exists but <code>compatible</code> is empty : not confirmed
OpenCL platform	<code>clinfo</code> not installed; no OpenCL ICD present
Userspace libraries	Missing: <code>libGL</code> , <code>libEGL</code> , <code>libGLESv2</code> , <code>libOpenCL</code> , <code>libvulkan</code> , <code>libgalcore</code>

Conclusion: Direct Rendering Manager (DRM) device nodes are present but the GPU is not accessible. The proprietary Vivante kernel driver is not loaded, the DT node is unconfirmed, and the NXP BSP userspace stack is absent. The GPU is **not usable** in the current system state.

2.5 Neural Processing Unit : 2.25 TOPS

The NPU (Vivante VIP / Neutron, 2.25 TOPS) has device nodes present but the driver has no hardware binding and the userspace inference stack is absent.

2.5.1 Neural Processing Unit (NPU)

Component	Status
NPU device nodes	/dev/galcore, /dev/vsi_daemon_ctrl : present
NPU kernel driver	galcore registered device nodes but has no hardware binding : no NPU probe in dmesg
NPU device tree	vipcore@38500000 compatible string probe returned empty : DT node not confirmed
NPU sysfs	/sys/bus/platform/drivers/galcore/ visible. Entries (gpu3DClockScale, gpu3DMinClock) are GPU-driver artefacts: the galcore instance here serves the GPU namespace, not the NPU; no dedicated NPU sysfs entries present.
Inference libraries	Missing: libneuralnetworks, libovxlib, libVSC, libCLC, TFLite delegate

Note: The NPU hardware is present on the i.MX8M Plus SoC, but the kernel driver is not properly bound to the device (no NPU probe in dmesg). The **galcore** driver seen in sysfs is actually handling the GPU (Vivante GC7000L), not the NPU (VeriSilicon VIP8000). No NPU userspace libraries are installed. **Conclusion:** Device nodes exist (driver registered) but without a confirmed DT node the driver cannot access the NPU hardware. The NPU is **not usable for inference** in the current state. The NXP BSP userspace stack is not apt-installable on a plain Debian rootfs.

2.6 Video Processing Unit : Hantro G1/G2

The VPU (H.264/H.265) is **not accessible**. No VPU codec device nodes are present. /dev/video0 and /dev/video1 are camera/ISP nodes, not codec nodes. No Hantro or Amphion kernel module is loaded. VPU Device Tree nodes are present (vpu_g1@38300000, vpu_g2@38310000, vpu_vc8000e@38320000) but no driver is bound to them, which makes the nodes unprobed.

Component	Status
VPU codec device nodes	/dev/mxc_vpu*, /dev/decoder* : absent
Camera/ISP nodes	/dev/video0, /dev/video1 : present (unrelated to VPU codec)
VPU kernel module	No Hantro/Amphion module loaded
VPU device tree	Nodes present (vpu_g1@38300000, vpu_g2@38310000, vpu_vc8000e@38320000) but no driver bound : unprobed
v4l2-ctl	Not installed

2.7 Performance Monitoring Unit (PMU)

Hardware performance counters are partially accessible.

Feature	Status
<code>perf_event_paranoid :2</code>	restricts kernel-context hardware counters; userspace-scoped counters (:u) unaffected
<code>perf stat (:u)</code>	Functional : 413 235 cycles:u, 137 873 instructions:u collected over 0.1s
<code>perf stat (kernel context)</code>	Blocked : cache-misses, branch-mispredictions unavailable at <code>paranoid=2</code>
PMU IRQ	arm-pmu IRQ 23, imx8_dds_perf_pmu IRQ 130
PMU event sources	armv8_cortex_a53, imx8_dds0, cs_etm, breakpoint, software, tracepoint, uprobe
Debugfs	Mounted (rw) but <code>arm_pmu</code> , <code>tracing</code> , <code>clk</code> subdirectories inaccessible : Permission denied

Note: Userspace-mode counters (:u) are functional without privilege. Full hardware counter access requires `perf_event_paranoid ≤ 1` or root.

2.8 WFE/SEV Cross-Core Synchronisation

All four cores are online (mask 0-3) and CPU affinity is controllable via `taskset` (confirmed: `core-pin OK`). WFE and SGX are not enumerated in `/proc/cpuinfo` flags : this is expected behaviour. WFE/SEV are mandatory ARMv8 instructions and are not advertised as optional CPU features; their absence from the flags field does not indicate absence from the hardware. `cyclicttest` is not installed so quantitative inter-core latency measurement was not performed.

2.9 System Readiness Summary

Accessible : functional without privilege

- CPU topology, frequency scaling, governor, cache hierarchy
- Memory and CMA pool (`/proc/meminfo`, `/proc/zoneinfo`)
- NEON/ASIMD : confirmed via `/proc/cpuinfo` flags on all four cores
- AES, SHA1/SHA2, CRC32 cryptographic extensions
- DRM device nodes (`/dev/dri/card0`, `renderD128`), only having a node presance.
- Userspace PMU counters (`perf stat :u : cycles, instructions`)
- PMU IRQ entries and event source devices
- `taskset` CPU affinity (all four cores pinnable)
- Software perf events (`perf list sw`)

Not Accessible : blocker and required action

Block	HW present	Blocker
GPU GC7000UL	Yes	Driver not loaded; DT node unconfirmed; NXP BSP userspace absent
NPU 2.25 TOPS	Yes	Driver has no HW binding; DT node unconfirmed; BSP inference stack absent
VPU H.264/H.265	Yes	No codec device nodes; Hantro driver absent; DT nodes present but unbound
PMU kernel-context counters	Yes	<code>perf_event_paranoid=2</code> : set to ≤ 1 or run as root
Debugfs (<code>arm_pmu</code> , <code>tracing</code> , <code>clk</code>)	Yes	Permission denied : requires root or <code>tracing</code> group membership
CPU hotplug control	Yes	Requires <code>CAP_SYS_ADMIN</code>
gcc / NEON compile test	N/A	gcc not installed : rootfs 91% full, install after freeing space
v4l-utils, rt-tests, clinfo	N/A	Not installed : rootfs 91% full, install after freeing space
NXP BSP stack (GPU/NPU/VPU)	N/A	Not apt-installable on plain Debian : requires NXP Yocto BSP rootfs

This concludes the compute and AI core assessment. The next chapters cover memory security, inter-core communication, and multimedia subsystems. this probing of the system was done via the script of `Compute.sh`, for the verification of the different aspects of this assesement.

Chapter 3

Memory & Security

This chapter describes the actual state of memory, error correction, cryptographic accelerators, security controllers, and one-time programmable fuses on the i.MX8M Plus platform based on runtime probing. Each subsection includes observed hardware accessibility and identifies gaps that must be addressed for full functionality.

3.1 Main Memory

The system uses LPDDR4/LPDDR4X. Runtime parameters and validation readiness are as follows.

Parameter	Observed value
MemTotal	1.6 GB (1627172 kB)
MemFree	1.2 GB (1262248 kB)
MemAvailable	1.4 GB (1429492 kB)
CMA total	448 MB (458752 kB) : ready for multimedia
DDR PMU event source	<code>imx8_ddr0</code> (confirmed under <code>/sys/bus/event_source/devices/</code>)
DDR PMU IRQ	IRQ 130 (<code>imx8_ddr_perf_pmu</code> , confirmed in <code>/proc/interrupts</code>)
Memory stress tools	<code>memtester</code> and <code>stress-ng</code> not installed

DDR performance / stress: The DDR PMU is present and can be used with `perf` to measure bandwidth and latency once `perf_event_paranoid` is relaxed. No memory stress test has been executed because `memtester` and `stress-ng` are missing.

3.2 On-Chip Memories

- **TCM (M7):** Not directly probed from Linux. Hardware supports 256 KB zero-wait-state SRAM (ITCM/DTCM). Requires M7 firmware to verify.
- **Secure RAM (CAAM):** 32 KB isolated SRAM inside CAAM engine. Kernel crypto algorithms are registered via ARMv8 CE extensions (see Section 3.6), but no CAAM-specific driver names appear and the `/dev/crypto` node is absent.

3.3 Memory Protection and Isolation

3.3.1 Resource Domain Controller (RDC)

Component	Status
Device tree node	Node address rdc@303d0000 referenced in script; compatible string returned no device tree node
Platform device	Not listed under /sys/bus/platform/devices/
Kernel module	No separate module (likely built into BSP kernel)
Debugfs	/sys/kernel/debug/rdc/ not accessible : permission denied (needs root)

Note: RDC is required to isolate 4 execution domains (A53, M7, DSP, etc.) across the different peripherals and the memory regions. The DT compatible string could not be confirmed. Without debugfs access, RDC configuration cannot be verified at runtime.

3.3.2 Central Security Unit (CSU)

The CSU grep returned no output : no DT compatible string confirmed by probe. No CSU platform device is listed under /sys/bus/platform/devices/ and no iomem entry was found. Access rights for bus masters are configured during boot via HAB or ATF; no runtime CSU control is available from userspace.

3.4 Inline ECC (DDR)

Check	Result
ECC device tree	grep returned no output : ECC DT node not confirmed
EDAC driver	No EDAC module loaded
EDAC sysfs	Present: /sys/devices/system/edac/mc/ directory exists; mc*/ce_count read fails : no memory controller instance populated
Boot message	EDAC MC: ECC not enabled (confirmed in dmesg)

Conclusion: ECC is disabled in the current boot configuration. The EDAC framework is present (mc/ directory visible) but unpopulated. To enable inline ECC, the DDR controller must be initialised with ECC mode (requires U-Boot and DRAM calibration changes) and the kernel must load `imx_ddr_edac`.

3.5 Hardware Semaphores (SEMA42)

SEMA42 is used for inter-core locking. Runtime detection shows:

- Device tree: grep returned no output : DT compatible string not confirmed by probe.

- Platform device: none visible under `/sys/bus/platform/devices/`.
- No kernel module loaded; driver is typically built into NXP BSP kernel.
- No iomem entry found for SEMA42.
- No userspace API present (requires NXP SEMA4 library).

3.6 Cryptographic Accelerator (CAAM)

CAAM provides AES, RSA, ECC, TRNG, and secure RAM. Current status:

Feature	Status
Device tree node	Compatible string search for "caam" returned no output : no DT node with "caam" in the compatible property found.
Platform devices	Present : <code>100000.caam-sm</code> , <code>30370000.caam-snvs</code> , <code>caam_secvio</code> (confirmed under <code>/sys/bus/platform/devices/</code>).
Kernel messages	<code>dmesg grep -i caam</code> returned no output : CAAM driver not initialised.
Loaded modules	No <code>caam_*</code> modules loaded (<code>lsmod</code>).
<code>/dev/crypto</code> node	Missing .
AF_ALG sockets	Blocked : <code>[Errno 97] Address family not supported</code> ; kernel config unreadable so exact cause unconfirmed, but behaviour confirms AF_ALG unavailable.

Note: CAAM hardware is recognised at the platform device level, but the driver is not loaded or built into the kernel. No initialisation messages appear in `dmesg`. This means cryptographic offload and hardware TRNG are not operational.

3.7 True Random Number Generator (TRNG)

Component	Status
<code>/dev/hwrng</code>	Present (character device <code>10:183</code> , root-owned: <code>crw---</code>). Only root can read.
Active RNG source	<code>/sys/class/misc/hw_random/rng_current</code> = <code>none</code> : no hardware source bound to <code>/dev/hwrng</code> .
Available RNG sources	<code>rng_available</code> empty. CAAM RNG driver (<code>caamrng</code>) not loaded. System relies solely on software entropy (<code>jitterentropy</code>).
Entropy available	256 bits (from kernel CRNG, not hardware).
TRNG kernel module	No <code>imx-rngc</code> or <code>caamrng</code> module loaded.
TRNG read test	Cannot be performed : - <code>xxd</code> missing (exit 127) : install <code>vim-common</code> .

- Even if `xxd` were present, `/dev/hwrng` is readable only by root (mode 600).
- Most importantly, with `rng_current = none`, reading `/dev/hwrng` would return no data regardless of permissions or tools.

Note: The system relies entirely on software entropy (jitterentropy). `/dev/hwrng` exists as a device node but no hardware source is bound to it (`rng_available` is empty). The TRNG read test could not be executed because `xxd` is not installed; this is a missing tool issue, not an access permission issue. Hardware TRNG will become available after loading `caamrng`.

3.8 DDR Performance Monitoring Unit (PMU)

The DDR PMU is confirmed as `imx8_ddr0` in `/sys/bus/event_source/devices/` and IRQ 130 is mapped in `/proc/interrupts`. Example usage once `perf_event_paranoid` is relaxed:

```
perf stat -e imx8_ddr0/cycles/ dd if=/dev/zero of=/dev/null bs=1M count=1000
```

No memory stress tools are installed and must be added for full validation.

3.9 On Chip One-Time Programmable Controller (OCOTP / eFuses)

Component	Status
nvmem device	<code>imx-ocotp0</code> present at <code>/sys/bus/nvmem/devices/imx-ocotp0/nvmem</code> , permissions <code>-rw-r-r-</code> (world-readable).
Raw read test (non-root)	ACCESSIBLE : <code>cat</code> returns binary data (non-zero length). Verified by reading 1536 bytes.
Hex dump test	Requires <code>xxd</code> (provided by <code>vim-common</code>). After installation, non-root can run <code>cat nvmem xxd head -4</code> to view fuse values.

Note: The hex dump test requires `xxd` (from `vim-common`). During initial testing `xxd` was missing; after installing it, non-root can run `cat nvmem | xxd | head -4` to view fuse values

3.10 Secure Non-Volatile Storage and Tamper Detection (SNVS)

Component	Status
Device tree	DT grep returned no output : compatible string not confirmed; platform devices visible at 30370000.snvs
Platform devices	30370000.caam-snvs, 30370000.snvs, snvs-lpgpr, snvs-powerkey, snvs-rtc-lp (all confirmed)
RTC	/dev/rtc0 and /dev/rtc1 present; snvs-rtc-lp registered as rtc0 (dmesg confirmed)
Tamper pins	No tamper DT nodes found by grep
External RTC	rtc-abx80x registered as rtc1; oscillator failure reported in dmesg
dmesg	SNVS RTC registered; system clock set to 1970-01-10 (RTC not battery-backed)

Note: The external RTC (abx80x, rtc1) reports an oscillator failure : its data is invalid. SNVS RTC (rtc0) is functional but was not backed by a battery at boot. Tamper detection requires explicit pin muxing and driver configuration, which is not enabled in the current DT.

3.11 Boot Security (HAB/AHAB)

No runtime check is available from Linux userspace. The chain-of-trust must be verified via U-Boot commands (`hab_status`). eFuses for SRK hash are not readable from this OS layer.

3.11.1 Summary Table of Memory & Security Readiness

Subsystem	Hardware present	Currently operational
Main DDR	Yes	Yes (no ECC)
RDC	Yes	Partial (no debug access; DT compatible string absent from probe)
CSU	Yes	Unknown (no runtime visibility; DT compatible string absent from probe)
Inline ECC	Optional	No (disabled; EDAC mc/present but empty)
SEMA42	Yes	No (DT compatible string absent; no platform device; no userspace API)

CAAM	Yes	No (DT compatible string absent; no dmesg; no module; /dev/crypto missing; AF_ALG compiled out of kernel — CONFIG_CRYPT_USER_API not set)
TRNG	Yes	No (no HW source registered; <code>rng_current</code> = none; caamrng not loaded)
OCOTP/eFuses	Yes	Partial (nvmem readable as non-root, 1536 bytes confirmed; hex dump blocked — <code>xxd</code> absent, use <code>od</code> as workaround)
SNVS / tamper	Yes	RTC works (rtc0); external RTC (rtc1) oscillator failure; tamper not configured
Boot security (HAB)	Yes	Cannot verify from Linux

Chapter 4

IPC & Inter-Core Communication

This chapter describes the actual state of inter-processor communication mechanisms between the A53, M7, and DSP domains on the i.MX8M Plus platform, based on runtime probing.

4.1 Messaging Unit (MU)

Hardware FIFO-based messaging between cores, used as transport for RPMsg.

Component	Observed status
Device tree nodes	DT grep returned no output via compatible string search; however, uevent confirms <code>OF_COMPATIBLE_0=fsl,imx8mp-mu(30aa0000.mailbox)</code> and <code>OF_COMPATIBLE_0=fsl,imx8-mu-dsp(30e60000.mailbox)</code> — DT nodes are present and bound.
Platform devices	Two mailbox devices confirmed: <code>30aa0000.mailbox</code> (A53M7), <code>30e60000.mailbox</code> (A53DSP); both bound to <code>DRIVER=imx_mu</code> .
iomem	Both devices appear in <code>/proc/iomem</code> with zeroed address ranges (<code>00000000-00000000</code>) — kernel registration confirmed, physical address mapping unresolved.
Kernel module	No separate module (built-in); <code>DRIVER=imx_mu</code> confirmed via uevent.
IRQ	Not confirmed — exact-word grep on <code>/proc/interrupts</code> returned no output. Substring false positives (<code>arm-pmu</code> , <code>imx8_ddr_perf_pmu</code>) were previously reported as hits. MU IRQ entries appear only after M7/DSP cores are started via <code>remoteproc</code> .
dmesg	<code>"imx mu driver is registered"</code> : confirmed.

Conclusion: MU driver is confirmed bound (`DRIVER=imx_mu`) to both mailbox devices, with DT compatible strings verified via uevent (`fsl,imx8mp-mu`, `fsl,imx8-mu-dsp`).

The earlier DT grep failure was a probe limitation, not an absence. iomem shows zeroed ranges — registration is confirmed but physical mapping is unresolved. No IRQ entries are present; these will appear after M7/DSP cores are started via remoteproc.

4.2 RPMsg / OpenAMP (A53 M7 messaging)

RPMsg provides virtio-based messaging; OpenAMP offers lifecycle management.

Component	Observed status
Device tree	DT compatible string search returned no output — RPMsg/remoteproc DT node not confirmed by probe.
Remoteproc devices	No remoteproc platform devices; /sys/class/remoteproc/ empty.
M7 firmware	Multiple ELF files confirmed in /lib/firmware/ (hello_world, rpmsg_lite_pingpong, str_echo, sai_low_power_audio).
RPMsg virtual devices	None : M7 not started.
RPMsg /dev nodes	None (/dev/rpmsg*, /dev/ttyRPMMSG* absent).
Kernel modules	No rpmsg/remoteproc/virtio modules loaded (likely built-in).
dmesg	"imx rpmsg driver is registered" : confirmed.

Note: The driver is registered (dmesg) and M7 firmware images are present, but the remoteproc class has no entries. This indicates either CONFIG_IMX_REMOTEPROC is not enabled in this kernel, or the M7 DT node is disabled. The RPMsg DT compatible string was not returned by the probe. libmetal-dev and libopen-amp-dev are **not available via apt** on this system (apt-cache show returned no output); userspace OpenAMP must be built from source.

4.3 FlexSPI / QSPI

FlexSPI controller for boot flash (QSPI NOR), supporting XIP and shared core access.

Component	Observed status
Device tree	Compatible string search returned no output : not confirmed by probe.
Platform device	Not listed under /sys/bus/platform/devices/.
iomem	grep -E '30bb0000' /proc/iomem returned no output : FlexSPI base address absent.
MTD devices	/proc/mtd present but contains header only : no partitions registered.
MTD block devices	None (/dev/mtd* absent).
SPI-NOR driver	No module loaded.
dmesg	No FlexSPI initialisation messages.

Conclusion: All five independent probes (DT compatible, platform device, iomem, dmesg, MTD) returned no output. SPI NOR is either not populated on this board or

the DT node is explicitly disabled. Without MTD devices, XIP and shared flash access are unavailable.

4.3.1 Summary Table of IPC Readiness

Subsystem	Hardware present	Currently operational
MU (A53M7)	Yes	Driver bound (<code>imx_mu</code>); DT confirmed via uevent (<code>fsl,imx8mp-mu</code>); iomem zeroed; no active IRQ channels
MU (A53DSP)	Yes	Driver bound (<code>imx_mu</code>); DT confirmed via uevent (<code>fsl,imx8-mu-dsp</code>); iomem zeroed; no active IRQ channels
RPMsg / OpenAMP	Yes (dmesg)	Driver registered; remoteproc class absent; M7 not started; libmetal/open-amp not in apt
FlexSPI / QSPI	Unconfirmed	All probes empty; SPI NOR likely unpopulated or DT node disabled

Chapter 5

Networking & TSN

5.1 Time-Sensitive Networking (TSN) and Ethernet

Component	Observed status
Ethernet interfaces	<code>ip</code> command missing (<code>iproute2</code> not installed); <code>eth0</code> confirmed present via <code>/sys/class/net/</code>
ENET_QOS (second GbE)	Confirmed absent — <code>eth0</code> only; no <code>eth1</code> visible via <code>sysfs</code>
PTP devices (IEEE 1588)	<code>ptp0</code> present under <code>30bf0000.ethernet</code> ; clock name: <code>stmmac ptp</code> ; caps file unreadable without root
TSN tc tool	<code>tc</code> (<code>iproute2</code>) not installed

5.2 Controller Area Network (FlexCAN)

Component	Observed status
FlexCAN DT node	Present and DT nodes confirmed (<code>can@308d0000</code> , <code>can@308c0000</code>)
CAN interfaces	INCONCLUSIVE — <code>ip link show type can</code> requires <code>iproute2</code> which is absent; hardware state unverifiable until <code>iproute2</code> and <code>can-utils</code> are installed

Chapter 6

Low-Speed Peripherals

6.1 Serial Peripheral Interfaces (eCSPI)

Component	Observed status
SPI device nodes	System shows multiple spi@30820000 nodes and ecspi compatibles

6.2 I2C Buses

Parameter	Observed status
I2C buses	2 of 6 present (/dev/i2c-0, /dev/i2c-1); both BLOCKED for non-root — mode 0600, owned root:root
i2cdetect	Not installed

Non-root access is blocked by permissions.

6.3 Pulse Width Modulation (PWM)

Component	Observed status
PWM chips	Confirmed MISSING — 0 pwmchip entries in /sys/class/pwm/; 0 PWM DT compatible matches

Chapter 7

Power & Clocks

7.1 Power Management (GPC)

Component	Observed status
GPC power domains (debugfs)	BLOCKED — /sys/kernel/debug/pm_genpd/ returns 0 entries for non-root; exit-0 caused false ACCESSIBLE classification in probe script
GPC power domain sysfs	ACCESSIBLE — 30 genpd entries confirmed under /sys/bus/genpd/devices/
DVFS (cpufreq)	Governor: ondemand; current 1.6 GHz, min 1.2 GHz, max 1.8 GHz

7.2 Clock System (CCM)

Component	Observed status
CCM clk_summary	BLOCKED — /sys/kernel/debug/clk/clk_summary returns 0 lines for non-root; /sys/kernel/debug/clk/ listing also empty

To unblock both CCM and GPC debugfs paths without a root session:

```
sudo chmod o+rx /sys/kernel/debug/{clk,pm_genpd}
```

Persist across reboots via /etc/rc.local or a systemd-tmpfiles rule. Alternatively:

```
sudo usermod -aG tracing $USER # re-login required
```

7.3 Timers and Watchdogs

Component	Observed status
-----------	-----------------

GPT / EPIT timers	DT nodes present (timer@302d0000, timer@302e0000, etc.) : enabled in device tree
POSIX hrtimer / timer_list	BLOCKED — /proc/timer_list returns 0 lines for non-root
RTC / SNVS HP	RTC0: snvs_rtc (30370000.snvs); RTC1: rtc-abx80x (I2C 1-0069); hwclock: Fri May 29 17:43:00 UTC 2026
Watchdog device nodes	ACCESSIBLE — /dev/watchdog, /dev/watchdog0, /dev/watchdog1 present and bound
Watchdog DT compatible	Present : multiple watchdog DT nodes confirmed (watchdog@30280000, watchdog@30290000, watchdog@302a0000) with compatible strings; driver is DT-driven

Chapter 8

Debug & Trace

8.1 Trace and Debug Interfaces (ETM / CoreSight)

Component	Observed status
CoreSight sysfs (/sys/bus/coresight/devices/)	Confirmed MISSING — 0 devices enumerated
CoreSight debugfs (/sys/kernel/debug/coresight/)	BLOCKED — directory traversable but returns 0 entries for non-root
ETM / CoreSight DT nodes	Confirmed MISSING — 0 matches for etm, coresight, tmc, funnel

8.2 Performance Counters (PMU)

Parameter	Observed status
perf_event_paranoid	2 — blocks kernel-space hardware counters for non-root
PMU event sources	armv8_cortex_a53, imx8_ddr0, cs_etm, breakpoint, software, tracepoint, uprobe
PMU IRQ	arm-pmu (GICv3 IRQ 23), imx8_ddr_perf_pmu (GICv3 IRQ 130)
perf stat (cycles)	PARTIAL — cycles:u and instructions:u confirmed; userspace-only scope under paranoid=2, not full hardware counter access
DSU cluster PMU	Not present — expected; Cortex-A53 cluster does not implement DSU PMU

8.3 Capability Summary

Subsystem	Status	Note
-----------	--------	------

MMC / uSDHC	ACCESSIBLE	mmcblk1 (SD 238.5 GB) + mmcblk2 (eMMC 29.1 GB)
PWM (×4)	MISSING	0 pwmchip entries, 0 DT entries — script false positive corrected
I2C buses	BLOCKED	Nodes present but mode 0600 root:root
GPC debugfs	BLOCKED	0 lines non-root; sysfs alternative returns 30 entries
CCM clk_summary	BLOCKED	0 lines non-root
hrtimer	BLOCKED	/proc/timer_list returns 0 lines non-root
CoreSight sysfs	MISSING	0 devices
CoreSight debugfs	BLOCKED	0 lines non-root
ETM DT nodes	MISSING	0 DT matches
perf stat	PARTIAL	cycles:u / instructions:u only; paranoid=2
PTP caps	BLOCKED	clock_name=stmmac ptp confirmed; caps unreadable non-root

Chapter 9

Suggested Solutions and Planification

This chapter provides concrete remediation steps for each subsystem identified as MISSING, BLOCKED, or INCONCLUSIVE in the preceding audit chapters (Compute & AI, Memory & Security, IPC, Networking & TSN, Low-Speed Peripherals, Power & Clocks, and Debug & Trace).

9.1 Userspace Fixes (No Reboot Required)

9.1.1 Free Disk Space First

The root filesystem is at 91 % capacity. Every subsequent step that calls `apt-get` will fail without available space.

```
sudo journalctl --vacuum-size=50M
sudo apt-get clean
sudo rm -rf /tmp/*
df -h /    # confirm headroom before proceeding
```

9.1.2 Install Missing Userspace Tools

A single `apt-get` pass resolves the majority of BLOCKED and INCONCLUSIVE states across all chapters.

```
sudo apt-get update
sudo apt-get install -y \
    iproute2          '# ip, tc - Networking/TSN/CAN' \
    can-utils          '# candump, cansend - FlexCAN' \
    i2c-tools          '# i2cdetect, i2cget - I2C' \
    ethtool            '# PTP, NIC diagnostics' \
    linuxptp           '# ptp4l, phc2sys - IEEE 1588' \
    v4l-utils          '# v4l2-ctl - VPU/ISP nodes' \
    alsa-utils         '# aplay - SAI/audio' \
    gpiod              '# gpiodetect - GPIO' \
    linux-perf         '# perf - PMU counters' \
    vim-common         '# xxd - OCOTP/TRNG hex dumps' \
    gcc               '# NEON compile test' \
    memtester          '# DDR stress' \
```

```
stress-ng          '# memory/CPU stress' \
rt-tests           '# cyclicttest - WFE/SEV latency'
```

9.1.3 Update perf_event_paranoid (PMU)

Currently at 2; full hardware counter access (cache-misses, branch-mispredictions, kernel-context) requires ≤ 1 .

```
# Temporary
sudo sysctl -w kernel.perf_event_paranoid=1
```

```
# Permanent
echo 'kernel.perf_event_paranoid = 1' | \
    sudo tee /etc/sysctl.d/99-perf.conf
sudo sysctl -p /etc/sysctl.d/99-perf.conf
```

Also add the user to the tracing group to unblock /sys/kernel/debug/tracing and arm_pmu debugfs:

```
sudo usermod -aG tracing $USER    # re-login required
```

9.1.4 Update I2C Node Permissions

/dev/i2c-0 and /dev/i2c-1 are root:root 0600.

```
sudo groupadd -f i2c
sudo usermod -aG i2c $USER
echo 'SUBSYSTEM=="i2c-dev", GROUP="i2c", MODE="0660"' | \
    sudo tee /etc/udev/rules.d/99-i2c.rules
sudo udevadm control --reload-rules && sudo udevadm trigger
```

9.1.5 Update /dev/hwrng Permissions (TRNG)

The TRNG node exists but is root-only (mode 0600). A udev rule grants group access now; hardware binding requires CAAM driver changes (Section 9.3.2).

```
echo 'KERNEL=="hwrng", GROUP="root", MODE="0644"' | \
    sudo tee /etc/udev/rules.d/99-hwrng.rules
sudo udevadm control --reload-rules && sudo udevadm trigger
```

9.1.6 Chnage the Debugfs Permissions (CCM, GPC, CoreSight, arm_pmu)

```
# Temporary
sudo chmod o+rx \
    /sys/kernel/debug/clk \
    /sys/kernel/debug/pm_genpd \
    /sys/kernel/debug/coresight \
    /sys/kernel/debug/tracing
```

```
# Permanent via tmpfiles.d
cat <<'EOF' | sudo tee /etc/tmpfiles.d/debugfs-perms.conf
z /sys/kernel/debug/clk          0755 root root -
z /sys/kernel/debug/pm_genpd     0755 root root -
z /sys/kernel/debug/coresight    0755 root root -
z /sys/kernel/debug/tracing      0755 root root -
EOF
```

Expected outcomes: `clk_summary` populated, GPC 30 power-domain entries readable, CoreSight topology visible, `arm_pmu` debugfs accessible.

9.2 CPU, PMU, and Memory Validation

9.2.1 NEON / ASIMD Compile Test

Once `gcc` is installed (Section 1.2), verify NEON at compile-time:

```
cat > /tmp/neon_test.c << 'EOF'
#include <arm_neon.h>
int main(void) {
    float32x4_t a = vdupq_n_f32(1.0f);
    float32x4_t b = vaddq_f32(a, a);
    return (int)vgetq_lane_f32(b, 0) - 2;
}
EOF
gcc -O2 -march=armv8-a+simd -o /tmp/neon_test /tmp/neon_test.c && \
    /tmp/neon_test && echo "NEON OK"
```

9.2.2 DDR Stress and PMU Bandwidth Measurement

```
# Stress test (install memtester first)
sudo memtester 512M 1

# DDR PMU bandwidth (requires perf_event Paranoid <= 1)
perf stat -e imx8_ddr0/cycles/ \
    dd if=/dev/zero of=/dev/null bs=1M count=1000
```

9.2.3 Inter-Core Latency (WFE/SEV)

```
# Requires rt-tests package
sudo cyclicttest --mlockall --smp -p99 -t4 -a0-3 -i200 -l5000
```

This confirms WFE/SEV cross-core synchronisation latency across all four A53 cores. No kernel or DT changes are required.

9.2.4 CPU Hotplug

CPU hotplug requires CAP_SYS_ADMIN:

```
sudo sh -c 'echo 0 > /sys/devices/system/cpu/cpu3/online'
sudo sh -c 'echo 1 > /sys/devices/system/cpu/cpu3/online'
```

9.3 Memory & Security Fixes

9.3.1 OCOTP / eFuse Hex Dump

The nvmem node is already world-readable. Install xxd (in vim-common) then read fuse values as non-root:

```
cat /sys/bus/nvmem/devices/imx-ocotp0/nvmem | xxd | head -16
# Workaround if xxd still absent:
cat /sys/bus/nvmem/devices/imx-ocotp0/nvmem | od -A x -t x1z | head -16
```

9.3.2 Enable CAAM Cryptographic Accelerator

CAAM platform devices are present but the driver is not loaded and CONFIG_CRYPT_USER_API is not set. Two steps are required:

1. **Kernel rebuild** — enable the following symbols:

```
CONFIG_CRYPT_DEV_FSL_CAAM=y
CONFIG_CRYPT_DEV_FSL_CAAM_JR=y
CONFIG_CRYPT_DEV_FSL_CAAM_CRYPT_API=y
CONFIG_CRYPT_DEV_FSL_CAAM_RNG=y
CONFIG_CRYPT_USER_API=y
CONFIG_CRYPT_USER_API_HASH=y
CONFIG_CRYPT_USER_API_SKCIPHER=y
```

2. **Load modules** (if built as m):

```
sudo modprobe caam_jr
sudo modprobe caamalg
sudo modprobe caamrng
# Verify
cat /sys/class/misc/hw_random/rng_current
ls /dev/crypto
```

9.3.3 Enable TRNG Hardware Source

After loading caamrng (above), bind it and verify:

```
echo caam-rng | sudo tee /sys/class/misc/hw_random/rng_current
cat /sys/class/misc/hw_random/rng_current      # should print: caam-rng
xxd /dev/hwrng | head -4                      # should return random bytes
```

9.3.4 Enable Inline ECC (DDR)

ECC is disabled at boot (EDAC MC: ECC not enabled in dmesg). Enabling it will allow the memory controller to automatically detect and correct data corruption (bit flips) in real-time and for that we it requires two coordinated changes:

1. **U-Boot / DRAM calibration:** set ECC mode in the DDR init sequence (board-specific; consult the board vendor).
2. **Kernel module:** ensure CONFIG_EDAC_IMX=y and load:

```
sudo modprobe imx_ddr_edac
# Verify
ls /sys/devices/system/edac/mc/mc0/
cat /sys/devices/system/edac/mc/mc0/ce_count
```

9.3.5 SNVS External RTC Oscillator Failure

The abx80x (rtc1) reports an oscillator failure. Replace or re-initialise the crystal, then force a time write to clear the fault flag:

```
sudo hwclock --systohc --rtc /dev/rtc1
dmesg | grep abx80x # oscillator failure flag should clear after write
```

9.4 IPC & Inter-Core Communication

9.4.1 Enable M7 Remoteproc (RPMsg)

The MU driver is bound, M7 firmware images are present in /lib/firmware/, but /sys/class/remoteproc/ is empty. The M7 DT node must be enabled.

```
/* In board .dts */
&rproc_m7 {
    memory-region = <&m7_reserved>, <&vdev0vring0>,
                  <&vdev0vring1>, <&vdev0buffer>;
    status = "okay";
};
```

Also verify kernel config:

```
zcat /proc/config.gz | grep -E \
'CONFIG_IMX_REMOTEPROC|CONFIG_RPMMSG|CONFIG_VIRTIO'
```

Required symbols: CONFIG_IMX_REMOTEPROC=y, CONFIG_RPMMSG_VIRTIO=y, CONFIG_VIRTIO=y.
After DTB rebuild and reboot, start the M7 core:

```
echo /lib/firmware/imx/rpmsg_lite_pingpong_rtos_linux_remote.elf | \
    sudo tee /sys/class/remoteproc/remoteproc0/firmware
echo start | sudo tee /sys/class/remoteproc/remoteproc0/state
ls /dev/rpmsg* # RPMsg endpoints should appear
```

For userspace OpenAMP, libmetal and libopen-amp are not available via apt on plain Debian and must be built from source:

```
git clone https://github.com/OpenAMP/libmetal.git
git clone https://github.com/OpenAMP/open-amp.git
# Follow CMake build instructions targeting aarch64-linux-gnu
```

9.4.2 FlexSPI / QSPI NOR

All five probes (DT, platform device, iomem, dmesg, MTD) returned empty. If SPI NOR is populated on the board, enable the DT node:

```
&flexspi {
    status = "okay";
    flash0: mt25qu512abb@0 {
        reg = <0>;
        compatible = "jedec,spi-nor";
        spi-max-frequency = <80000000>;
        spi-tx-bus-width = <4>;
        spi-rx-bus-width = <4>;
    };
};
```

Verify after reboot:

```
cat /proc/mtd
ls /dev/mtd*
```

9.5 Kernel Configuration Changes (Recompile Required)

The table below consolidates all Kconfig symbols identified as missing or unconfirmed across all chapters. Verify current state first:

```
zcat /proc/config.gz | grep -E \
    'CONFIG_CAN|CONFIG_SPI|CONFIG_PWM_IMX|CONFIG_CORESIGHT|\
CONFIG_PTP|CONFIG_RPMMSG|CONFIG_IMX_REMOTEPROC|\
CONFIG_CRYPTODEV_FSL|CONFIG_CRYPTODEV_USER|\
CONFIG_EDAC_IMX|CONFIG_VIRTIO'
```

Symbol	Subsystem
CONFIG_CAN	FlexCAN
CONFIG_CAN_FLEXCAN	FlexCAN
CONFIG_SPI_IMX	eCSPI
CONFIG_SPI_SPIDEV	spidev nodes
CONFIG_PWM_IMX27	PWM1: 4
CONFIG_I2C_IMX	I2C2: 5
CONFIG_CORESIGHT	CoreSight

CONFIG_CORESIGHT_LINKS_AND_SINKS	TMC / Funnel
CONFIG_CORESIGHT_SOURCE_ETM4X	ETM
CONFIG_PTP_1588_CLOCK	PTP caps
CONFIG_FEC	ENET_QOS
CONFIG_IMX_REMOTEPROC	M7 re-moteproc
CONFIG_RPMSG_VIRTIO	RPMsg
CONFIG_VIRTIO	RPMsg transport
CONFIG_CRYPTODEV_FSL_CAAM	CAAM
CONFIG_CRYPTODEV_FSL_CAAM_RNG	TRNG
CONFIG_CRYPTODEV_USER_API	AF_ALG / /dev/crypto
CONFIG_EDAC_IMX	DDR ECC

9.6 Device Tree Modifications

All nodes below are absent or unbound in the current DTB. Edit the board `.dts` / `.dtsi` and rebuild with `make dtbs`.

9.6.1 PWM1: 4

```
&pwm1 { pinctrl-names = "default"; pinctrl-0 = <&pinctrl_pwm1>;
        status = "okay"; };
&pwm2 { pinctrl-names = "default"; pinctrl-0 = <&pinctrl_pwm2>;
        status = "okay"; };
&pwm3 { pinctrl-names = "default"; pinctrl-0 = <&pinctrl_pwm3>;
        status = "okay"; };
&pwm4 { pinctrl-names = "default"; pinctrl-0 = <&pinctrl_pwm4>;
        status = "okay"; };
```

```
ls /sys/class/pwm/
```

9.6.2 Additional I2C Buses (I2C2: I2C5)

```
&i2c2 { clock-frequency = <400000>; pinctrl-names = "default";
        pinctrl-0 = <&pinctrl_i2c2>; status = "okay"; };
/* Repeat for i2c3, i2c4, i2c5 */
```

```
ls /dev/i2c-*    # expect i2c-0 through i2c-5
```

9.6.3 Second Ethernet — ENET_QOS (eth1) and TSN

```
&eqos {
    pinctrl-names = "default";
    pinctrl-0 = <&pinctrl_eqos>;
    phy-mode = "rgmii-id";
```

```

phy-handle = <&ethphy1>;
status = "okay";
mdio {
    #address-cells = <1>;
    #size-cells = <0>;
    ethphy1: ethernet-phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <1>;
    };
};
};

```

After reboot, configure TSN Credit-Based Shaper (CBS) using `tc` (installed via `iproute2`):

```

ip link show eth1
# CBS example on egress queue 0
tc qdisc add dev eth1 root handle 1: mqprio \
    num_tc 3 map 2 2 1 0 2 2 2 2 2 2 2 2 2 2 2 2 \
    queues 1@0 1@1 2@2 hw 0
tc qdisc replace dev eth1 parent 1:1 cbs \
    idleslope 20000 sendslope -980000 \
    hicredit 10 locredit -50 offload 1

```

9.6.4 CoreSight / ETM

```

etm0: etm@28040000 {
    compatible = "arm,coresight-etm4x", "arm,primecell";
    reg = <0 0x28040000 0 0x1000>;
    cpu = <&cpu0>;
    clocks = <&clk IMX8MP_SYSTEM_PLL1_CLK>;
    clock-names = "apb_pclk";
    out-ports { port { etm0_out: endpoint {
        remote-endpoint = <&funnel_in0>; }; }; };
};
/* Repeat etm1: etm3 for cpu1: cpu3 */
funnel@28041000 {
    compatible = "arm,coresight-static-funnel";
    /* wire in-ports to etm0_out .. etm3_out,
       out-port to TMC/ETB */
};

```

Full base addresses are in the i.MX8M Plus Reference Manual, *CoreSight Debug Architecture* section. After rebuild:

```
ls /sys/bus/coresight/devices/
```

9.6.5 VPU — Bind Hantro Driver to Existing DT Nodes

VPU DT nodes (`vpu_g1@38300000`, `vpu_g2@38310000`, `vpu_vc8000e@38320000`) are present but no driver is bound. Enable the upstream Hantro driver:

```
CONFIG_VIDEO_HANTRO=y
CONFIG_VIDEO_HANTRO_HEVC=y
```

Install V4L2 tools after kernel rebuild:

```
v4l2-ctl --list-devices
ls /dev/video*
```

9.7 BSP Migration (GPU, NPU, Full Multimedia)

The GPU (GC7000UL), NPU (VeriSilicon VIP8000, 2.25 TOPS), HiFi4 DSP, and ISP require proprietary firmware and out-of-tree drivers that are not available in the upstream kernel or plain Debian.

9.7.1 Option A — NXP Yocto BSP

```
mkdir imx-yocto && cd imx-yocto
repo init -u https://github.com/nxp-imx/imx-manifest \
    -b imx-linux-scarthgap \
    -m imx-6.6.3-1.0.0.xml
repo sync
DISTRO=fsl-imx-xwayland MACHINE=imx8mpevk \
    source imx-setup-release.sh -b build-imx8mp
bitbake imx-image-multimedia
```

Provides: galcore GPU driver, galcore NPU binding, Hantro VPU, ISP firmware, HiFi4 DSP firmware, complete BSP device trees. Customise MACHINE to match the target board.

9.7.2 Option B — NXP Debian Overlay (Partial)

```
# Add NXP Debian repository (check availability for bookworm)
echo "deb [trusted=yes] https://www.nxp.com/lgfiles/NMG/MAD/YOCTO/ \
    bookworm main" | sudo tee /etc/apt/sources.list.d/nxp.list
sudo apt-get update
sudo apt-get install imx-gpu-viv imx-vpu-hantro \
    imx8-isp firmware-imx
```

9.8 Full Remediation Priority Table

#	Subsystem	Action
1	Disk space	<pre>sudo journalctl -vacuum-size=50M sudo apt-get clean sudo rm -rf /tmp/* Verify with df -h /</pre>

- 2 Userspace tools `sudo apt-get update`
`sudo apt-get install -y iproute2 gcc perf`
`vim-common memtester stress-ng rt-tests`
`i2c-tools can-utils v4l-utils alsa-utils gpiod`
`linuxptp ethtool`
- 3 `perf_event_paranoid` Temporary: `sudo sysctl -w`
`kernel.perf_event_paranoid=1`
Permanent:
`echo 'kernel.perf_event_paranoid = 1' | sudo`
`tee /etc/sysctl.d/99-perf.conf`
`sudo sysctl -p /etc/sysctl.d/99-perf.conf`
- 4 I2C permissions Create udev rule:
`echo 'SUBSYSTEM=="i2c-dev",`
`GROUP="i2c", MODE="0660"' | sudo tee`
`/etc/udev/rules.d/99-i2c.rules`
`sudo groupadd -f i2c`
`sudo usermod -aG i2c $USER`
`sudo udevadm control -reload-rules && sudo`
`udevadm trigger`
- 5 `/dev/hwrng` permis- `echo 'KERNEL=="hwrng", GROUP="root",`
sions `MODE="0644"' | sudo tee`
`/etc/udev/rules.d/99-hwrng.rules`
`sudo udevadm control -reload-rules && sudo`
`udevadm trigger`
- 6 Debugfs (CCM, GPC, Temporary:
tracing) `sudo chmod o+rx /sys/kernel/debug/clock`
`/sys/kernel/debug/pm_genpd`
`/sys/kernel/debug/coresight`
`/sys/kernel/debug/tracing`
Permanent via `/etc/tmpfiles.d/debugfs-perms.conf`:
`z /sys/kernel/debug/clock 0755 root root - etc.`
- 7 OCOTP hex dump Install vim-common for xxd:
`sudo apt-get install vim-common`
Read fuses:
`cat /sys/bus/nvmem/devices/imx-ocotp0/nvmem |`
`xxd | head -16`
Workaround without xxd: use `od -A x -t x1z`
- 8 WFE/SEV latency Install rt-tests:
measurement `sudo apt-get install rt-tests`
Run cyclicttest on all four cores:
`sudo cyclicttest -mlockall -smp -p99 -t4 -a0-3`
`-i200 -l5000`

- 9 NEON compile test Install gcc, then compile and test:
 cat > /tmp/neon_test.c « 'EOF'
 #include <arm_neon.h>
 int main(void) { float32x4_t a =
 vdupq_n_f32(1.0f); return 0; }
 EOF
 gcc -O2 -march=armv8-a+simd -o /tmp/neon_test
 /tmp/neon_test.c && /tmp/neon_test
- 10 DDR stress / PMU Install memtester:
 bandwidth sudo apt-get install memtester
 Run memory stress: sudo memtester 512M 1
 Measure DDR bandwidth (requires
 perf_event_paranoid=1):
 perf stat -e imx8_ddr0/cycles/ dd if=/dev/zero
 of=/dev/null bs=1M count=1000
- 11 SNVS RTC oscillator Write system time to external RTC (rtc1) to clear oscilla-
 tor failure flag:
 sudo hwclock -systohc -rtc /dev/rtc1
 Check dmesg | grep abx80x to confirm flag cleared
- 12 CAAM / TRNG Kernel rebuild required. Enable these symbols in kernel
 driver config:
 CONFIG_CRYPTODEV_FSL_CAAM=y,
 CONFIG_CRYPTODEV_FSL_CAAM_RNG=y,
 CONFIG_CRYPTODEV_USER_API=y,
 CONFIG_CRYPTODEV_USER_API_SKCIPHER=y
 After boot, load modules (if built as m):
 sudo modprobe caam_jr caamalg caamrng
 Bind TRNG: echo caam-rng | sudo tee
 /sys/class/misc/hw_random/rng_current
- 13 DDR ECC Enable ECC in U-Boot DDR initialization (board-
 specific).
 Set kernel config CONFIG_EDAC_IMX=y and load module:
 sudo modprobe imx_ddr_edac
 Verify: ls /sys/devices/system/edac/mc/mc0/ and
 read ce_count
- 14 FlexCAN (CAN bus) DT nodes already present (verified). Enable kernel config:
 CONFIG_CAN=y, CONFIG_CAN_FLEXCAN=y
 Install can-utils (already in step 2). After reboot,
 configure CAN interface:
 sudo ip link set can0 up type can bitrate
 500000
 Test with candump can0
- 15 eCSPI / spidev DT nodes already present (verified). Enable kernel config:
 CONFIG_SPI_IMX=y, CONFIG_SPI_SPIDEV=y
 Rebuild DTB only if nodes have status = "disabled".
 After reboot, ls /dev/spidev* should show nodes.
 Optionally add udev rule for permissions.

- 16 PWM1: 4 Enable DT nodes (set `status = "okay"`) and `pinctrl`.
 Kernel config: `CONFIG_PWM_IMX27=y`
 Rebuild DTB. After reboot:
 `ls /sys/class/pwm/` should show `pwmchip0` to `pwmchip3`
- 17 I2C2: 5 Enable additional I2C buses in DT (`pinctrl`, `clock-frequency`, `status`).
 Kernel config: `CONFIG_I2C_IMX=y` (usually built-in).
 Rebuild DTB. After reboot:
 `ls /dev/i2c-*` should show `i2c-0` through `i2c-5`
- 18 ENET_QOS / TSN Add DT node for `eqos` with PHY configuration (`mdio`, `phy-handle`).
 Ensure `CONFIG_FEC=y` and `CONFIG_PTP_1588_CLOCK=y`.
 Rebuild DTB. Install `iproute2` (step 2).
 After reboot, configure TSN Credit-Based Shaper with `tc`.
- 19 M7 remoteproc / Enable `&rproc_m7` in DT, set `status = "okay"`.
 RPMsg Kernel config: `CONFIG_IMX_REMOTEPROC=y`,
 `CONFIG_RPMMSG_VIRTIO=y`, `CONFIG_VIRTIO=y`.
 Rebuild kernel and DTB. After reboot, start firmware:
 `echo ... | sudo tee /sys/class/remoteproc/remoteproc0/firmware`
 `echo start | sudo tee /sys/class/remoteproc/remoteproc0/state`
- 20 FlexSPI / QSPI NOR If SPI NOR flash is populated, add DT node for `flexspi`
 with `status = "okay"`.
 Set `CONFIG_SPI_NXP_FLEXSPI=y` (or
 `CONFIG_SPI_IMX_FLEXSPI`).
 Rebuild DTB. After reboot, check `cat /proc/mtd` and
 `ls /dev/mtd*`
- 21 VPU Hantro binding DT nodes already present (`vpu_g1`, `vpu_g2`,
 `vpu_vc8000e`).
 Kernel config: `CONFIG_VIDEO_HANTRO=y` and
 `CONFIG_VIDEO_HANTRO_HEVC=y`.
 Rebuild kernel. Install `v4l-utils` (step 2). After reboot:
 `v4l2-ctl -list-devices` should show Hantro devices.
- 22 CoreSight / ETM Add full CoreSight topology to DT (ETM4x for each
 CPU, funnel, TMC/ETF, replicator).
 Enable kernel config: `CONFIG_CORESIGHT=y`,
 `CONFIG_CORESIGHT_SOURCE_ETM4X=y`,
 `CONFIG_CORESIGHT_LINKS_AND_SINKS=y`,
 `CONFIG_CORESIGHT_CPU_DEBUG=y`.
 Rebuild kernel and DTB. After reboot:
 `ls /sys/bus/coresight/devices/` should show ETM,
 funnel, etc.

- 24 GPU / NPU / DSP / ISP Full NXP Yocto BSP migration required (proprietary out-of-tree drivers).
Use NXP manifest for scarthgap:
`repo init -u https://github.com/nxp-imx/imx-manifest
-b imx-linux-scarthgap -m imx-6.6.3-1.0.0.xml
DISTRO=fsl-imx-xwayland MACHINE=imx8mpevk
source imx-setup-release.sh -b build-imx8mp
bitbake imx-image-multimedia`
Alternative Debian overlay (if available) but Yocto is recommended.
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Appendix A

Acronyms and Component Definitions

Acronym	Full Name	Role on i.MX8M Plus
AHAB	Advanced High Assurance Boot	Second-generation NXP secure boot framework; authenticates bootloader and firmware images using signed containers.
ASIMD	Advanced SIMD	ARMv8 name for NEON; 128-bit SIMD register file used for vector integer and floating-point operations.
ASRC	Asynchronous Sample Rate Converter	Hardware block that resamples audio streams between clock domains without CPU intervention.
ATF	ARM Trusted Firmware	Open-source reference implementation of the ARM Trusted Board Boot Requirements; runs in EL3 and configures security settings before Linux boots.
BSP	Board Support Package	Vendor-supplied collection of kernel patches, device tree files, drivers, and userspace libraries that enable hardware-specific features.
CAAM	Cryptographic Acceleration and Assurance Module	NXP hardware security engine providing AES, RSA, ECC, TRNG, and 32 KB of isolated secure RAM.
CCM	Clock Controller Module	Central clock generation and gating block; controls root clocks, dividers, and peripheral clock enables across the SoC.
CMA	Contiguous Memory Allocator	Linux kernel mechanism that reserves a physically contiguous memory region for DMA-capable devices such as the ISP and VPU.

CoreSight	CoreSight Debug and Trace	ARM on-chip debug and trace architecture; comprises ETM, TMC, funnel, and STM components connected via the ATB bus.
CRC	Cyclic Redundancy Check	ARMv8 hardware instruction set extension (<code>crc32</code>) for fast polynomial-based data integrity checks.
CSU	Central Security Unit	Configures bus-master access rights to peripherals; determines whether each master can access a given slave in secure or non-secure mode.
DRAM	Dynamic Random-Access Memory	Volatile main memory; on this platform LPDDR4/LPDDR4X at up to 1.6 GB.
DRM	Direct Rendering Manager	Linux kernel subsystem that manages GPU and display hardware; exposes <code>/dev/dri/card*</code> and <code>renderD*</code> device nodes.
DSP	Digital Signal Processor	HiFi4 DSP core on the i.MX8M Plus; intended for audio pre/post-processing offload from the A53 cluster.
DT	Device Tree	Hardware description data structure passed by the bootloader to the Linux kernel; defines peripherals, addresses, IRQs, and clocks without hardcoding them in the kernel.
DTCM	Data Tightly Coupled Memory	Zero-wait-state SRAM directly connected to the M7 data bus; used for latency-critical data buffers.
DVFS	Dynamic Voltage and Frequency Scaling	Power management technique that adjusts CPU voltage and frequency at run-time based on load; controlled via the <code>cpufreq</code> subsystem.
ECC	Error-Correcting Code	Hardware mechanism in the DDR controller that detects and corrects single-bit memory errors; requires specific U-Boot and kernel configuration.
EDAC	Error Detection and Correction	Linux kernel framework that exposes memory error counters and status via <code>sysfs (/sys/devices/system/edac/)</code> .
eCSPI	Enhanced Configurable Serial Peripheral Interface	NXP SPI master/slave controller; supports up to 4 chip selects and transfer widths from 1 to 32 bits.

eMMC	embedded MultiMediaCard	On-board NAND flash storage with a standardised MMC interface; appears as <code>mmcblk2</code> on this system (29.1 GB).
ENET_QOS	Ethernet with Quality of Service	Second Gigabit Ethernet controller on the i.MX8M Plus; implements IEEE 802.1Q VLAN and TSN credit-based shaping.
EPIT	Enhanced Periodic Interrupt Timer	General-purpose 32-bit timer peripheral; not enabled in the current device tree.
ETM	Embedded Trace Macrocell	Per-core instruction trace unit; records the program execution stream for offline analysis via CoreSight.
FlexCAN	Flexible Controller Area Network	NXP CAN 2.0B / CAN FD controller; used for real-time embedded bus communication; DT node absent on this board.
FlexSPI	Flexible Serial Peripheral Interface	High-speed SPI controller supporting single/dual/quad/octal modes; used to boot from QSPI NOR flash and supports XIP.
GC7000UL	Graphics Core 7000 Ultra-Lite	Vivante GPU integrated into the i.MX8M Plus; supports OpenGL ES 3.1, Vulkan 1.1, and OpenCL 1.2; requires NXP proprietary driver stack.
GPC	General Power Controller	Manages power domain sequencing, wake-up sources, and low-power state transitions for the SoC power islands.
GPT	General Purpose Timer	32-bit timer with input capture and output compare; not enabled in the current device tree.
GPU	Graphics Processing Unit	Programmable processor for 2D/3D rendering and general-purpose compute; see GC7000UL.
HAB	High Assurance Boot	NXP secure boot mechanism for i.MX8M; validates signed images using public key hashes stored in eFuses.
IPC	Inter-Processor Communication	Generic term for any mechanism that transfers data or control signals between separate processor cores (A53, M7, DSP).

IRQ	Interrupt Request	Asynchronous signal from a peripheral to the CPU requesting service; managed by the GICv3 interrupt controller.
ISP	Image Signal Processor	Hardware pipeline for raw camera sensor data processing (demosaic, noise reduction, colour correction).
ITCM	Instruction Tightly Coupled Memory	Zero-wait-state SRAM on the M7 instruction bus; used to store time-critical firmware code.
LPDDR4	Low Power Double Data Rate 4	Mobile DRAM standard; lower power consumption than standard DDR4 at the cost of reduced capacity per die.
LCDIF	LCD Interface	Parallel RGB display controller; drives flat-panel displays directly or feeds into MIPI DSI bridge; DT node absent on this system.
MIPI CSI-2	Mobile Industry Processor Interface Camera Serial Interface 2	Serial interface standard for connecting image sensors to the ISP; DT node absent on this system.
MIPI DSI	Mobile Industry Processor Interface Display Serial Interface	High-speed serial interface for connecting the SoC to DSI display panels; DT node absent on this system.
MTD	Memory Technology Device	Linux subsystem for raw flash devices (NOR, NAND); exposes <code>/dev/mtd*</code> nodes and <code>/proc/mtd</code> .
MU	Messaging Unit	Hardware FIFO register block providing interrupt-driven message passing between A53, M7, and DSP cores; transport layer for RPMsg.
NEON	(no expansion; ARM marketing name)	ARM SIMD extension implementing 128-bit vector operations; exposed as <code>asimd</code> in <code>/proc/cpuinfo</code> on AArch64.
NPU	Neural Processing Unit	Vivante VIP / Neutron fixed-function inference accelerator; rated at 2.25 TOPS; requires NXP BSP driver and inference libraries.
NUMA	Non-Uniform Memory Access	Memory architecture where access latency depends on which processor accesses which memory bank; this platform has a single NUMA node.
OCOTP	On-Chip One-Time Programmable Controller	Manages the eFuse array storing device-unique identifiers, MAC addresses, SRK hashes, and boot configuration bits.

OpenAMP	Open Source Asymmetric Multi-Processing	Framework providing lifecycle management and RPMsg-based IPC for heterogeneous multi-core systems; userspace library.
PMP	Physical Memory Protection	RISC-V term; on ARM the equivalent is MPU. Not directly applicable here but sometimes confused with PMU.
PMU	Performance Monitoring Unit	Hardware counters in the CPU (and DDR controller) that record events such as cycles, instructions retired, and cache misses; accessed via the <code>perf</code> subsystem.
PTP	Precision Time Protocol	IEEE 1588 network time synchronisation protocol; hardware timestamping is provided by the <code>stmmac</code> Ethernet controller (<code>ptp0</code>).
PXP	Pixel Pipeline	2D graphics accelerator for image format conversion, rotation, and scaling; used in display pipelines; DT node absent on this system.
PWM	Pulse Width Modulation	Generates rectangular waveforms with controllable duty cycle; used for fan control, backlight, and motor drive; all four instances absent from the current DT.
QSPI	Quad Serial Peripheral Interface	SPI operating with four data lines simultaneously; used with NOR flash for faster reads and XIP capability.
RDC	Resource Domain Controller	Enforces memory and peripheral access isolation between the four execution domains (A53, M7, DSP, and a fourth domain) at the bus fabric level.
RPMsg	Remote Processor Messaging	Virtio-based inter-core messaging protocol layered on top of the MU hardware; provides named endpoints and a standard Linux kernel API.
RTC	Real-Time Clock	Battery-backed counter that maintains wall-clock time across power cycles; <code>rtc0</code> is the SNVS internal RTC, <code>rtc1</code> is the external abx80x.
SAI	Synchronous Audio Interface	NXP I2S-compatible audio serial interface for connecting codecs and other digital audio peripherals.

SDMA	Smart Direct Memory Access	Programmable DMA controller running microcode scripts; SDMA3 is the instance serving audio peripherals on the i.MX8M Plus.
SEMA42	Semaphore Unit 42	Hardware semaphore block providing atomic test-and-set operations for mutual exclusion between processor cores.
SIMD	Single Instruction Multiple Data	Processor execution model that applies one instruction to multiple data elements in parallel; implemented by NEON/ASIMD on this platform.
SNVS	Secure Non-Volatile Storage	Always-on domain containing the secure RTC, tamper detection logic, and a small set of non-volatile registers maintained across power cycles.
SRK	Super Root Key	Set of public keys whose hashes are burned into eFuses; used by HAB/AHAB to authenticate the bootloader certificate chain.
SVE	Scalable Vector Extension	ARM vector ISA extension introduced in ARMv8.2 with variable vector lengths (128: 2048 bit); not present on Cortex-A53 (ARMv8.0).
TCM	Tightly Coupled Memory	Low-latency SRAM directly connected to the M7 processor buses, bypassing the cache hierarchy; split into ITCM (instruction) and DTCM (data).
TMC	Trace Memory Controller	CoreSight component that buffers ETM trace data to on-chip SRAM (ETB mode) or off-chip DDR (ETR mode).
TOPS	Tera Operations Per Second	Unit for neural network inference throughput; the NPU on this platform is rated at 2.25 TOPS.
TRNG	True Random Number Generator	Hardware entropy source inside CAAM that samples analog noise; feeds <code>/dev/hwrng</code> when the <code>caamrng</code> driver is loaded.
TSN	Time-Sensitive Networking	Set of IEEE 802.1 standards (CBS, TAS, PSFP) enabling deterministic latency over Ethernet; supported in hardware by the ENET_QOS controller.
uSDHC	Ultra Secured Digital Host Controller	NXP SD/MMC host controller; two instances are active: one for the SD card (<code>mmcbl1k1</code>), one for eMMC (<code>mmcbl1k2</code>).

VPU	Video Processing Unit	Hantro G1/G2 hardware video codec engine; decodes H.264 (G1) and H.265/VP9 (G2) and encodes H.264 (VC8000E); driver unbound on this system.
WFE / SEV	Wait For Event / Send Event	Mandatory ARMv8 instructions for low-power cross-core synchronisation; not enumerated in <code>/proc/cpuinfo</code> flags because they are architectural requirements, not optional features.
XIP	Execute In Place	Executing code directly from NOR flash without copying it to RAM first; requires FlexSPI controller and a memory-mapped flash device.
